

Pathology of non-neoplastic skin diseases

Principles and Practices of Osteopathic Medicine I

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Session Objectives

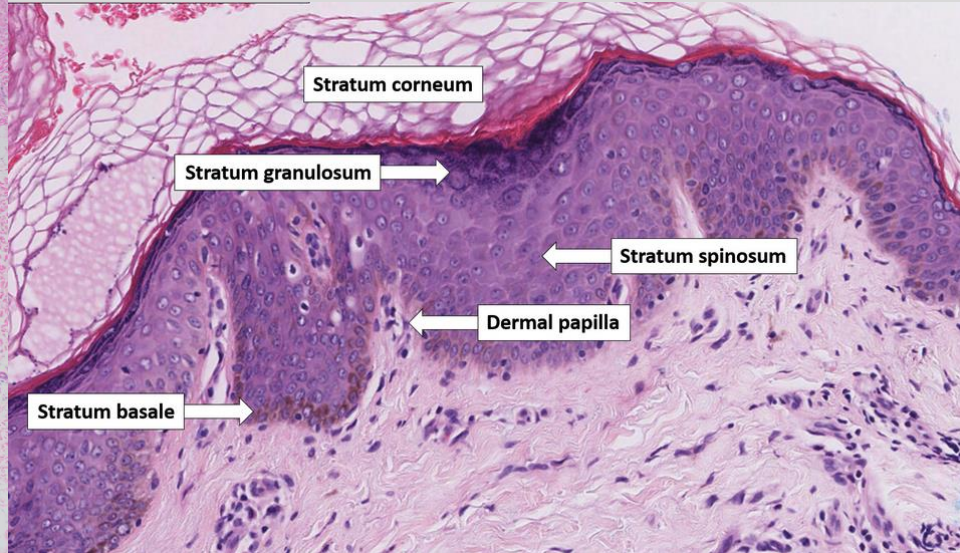
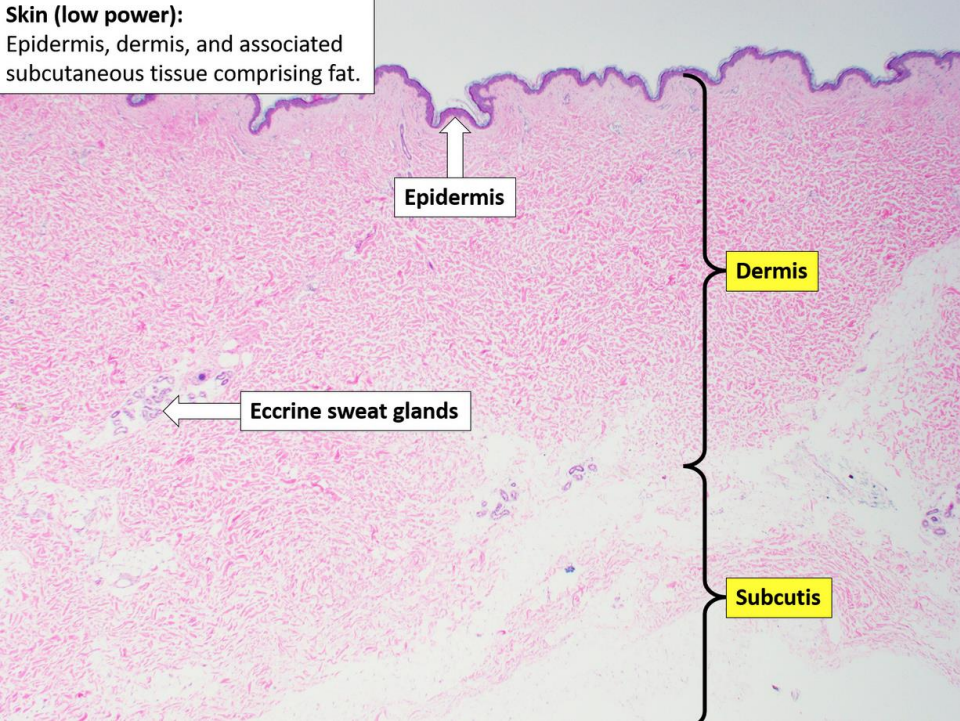
1. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of **acute inflammatory** dermatoses which include **urticaria, acute eczematous dermatitis, erythema multiforme, Stevens Johnson syndrome**
2. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of **chronic inflammatory** dermatoses which include **psoriasis** and **lichen planus**
3. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic), and major clinical correlations of major **bullous** diseases which include **pemphigus vulgaris, bullous pemphigoid, and dermatitis herpetiformis**
4. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic) and major clinical correlations of **acne vulgaris and rosacea**
5. Describe, discuss, and explain the pathogenesis, morphological features (gross and microscopic) and major clinical correlations of **erythema nodosum (panniculitis)**

Skin Histology Review

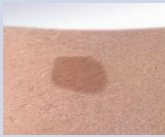
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Skin (low power):
Epidermis, dermis, and associated
subcutaneous tissue comprising fat.



Skin pathology terminology

Macroscopic Feature	Description	Image
Macule/Patch	Well circumscribed, flat lesion distinguished from surrounding skin by color. Macules <5mm, patches >5mm	
Papule/Nodule	Elevated dome-shaped or flat-topped lesion. Papules <5mm, nodules >5mm	
Plaque	Elevated flat-topped lesion, usually >5mm (can be coalesced papules)	
Pustule	Discrete, pus-filled, raised lesion	
Vesicle/Bulla/Blister	Fluid-filled raised lesion. Vesicles <5mm, bulla >5mm (blister general term for both)	
Wheal	Itchy, transient, elevated lesions with variable blanching and erythema formed as the result of dermal edema	

More skin pathology terminology

Microscopic Feature	Description
Acanthosis	Diffuse epidermal hyperplasia
Dyskeratosis	Abnormal, premature keratinization within cells below the stratum granulosum
Hyperkeratosis	Thickening of the stratum corneum, often associated with a qualitative abnormality of the keratin
Lentiginous	Linear pattern of melanocyte proliferation within the epidermal basal cell layer
Parakeratosis	Keratinization with retained nuclei in the stratum corneum
Spongiosis	Intracellular edema of the epidermis

Review of hypersensitivity reactions

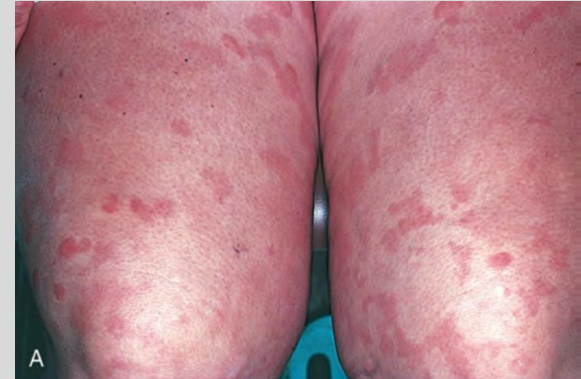
- **Type I** (immediate) – **IgE** binds and activates mast cells (ex. anaphylaxis, asthma, urticaria)
- **Type II** (cytotoxic) – **IgG/IgM** antibody dependent, bind to antigen (complement, opsonization/phagocytosis, natural killer (NK) cells bind Ab Fc region, cell dysfunction – hyperthyroidism in Graves' disease, etc.), other examples: Incompatible RBC transfusion reaction, Goodpasture syndrome
- **Type III** (immune complex) – **Ag-Ab complexes** deposition in tissues causing inflammation (ex. lupus, rheumatoid arthritis)
- **Type IV** (delayed type) – cell mediated toxicity, **T cell** mediated, 24-72 hours to develop, antibody independent reactions (ex. contact dermatitis, tuberculin skin test)

Acute inflammatory dermatoses

1. Urticaria
2. Acute eczematous dermatitis
3. Erythema multiforme

Urticaria (hives): Overview

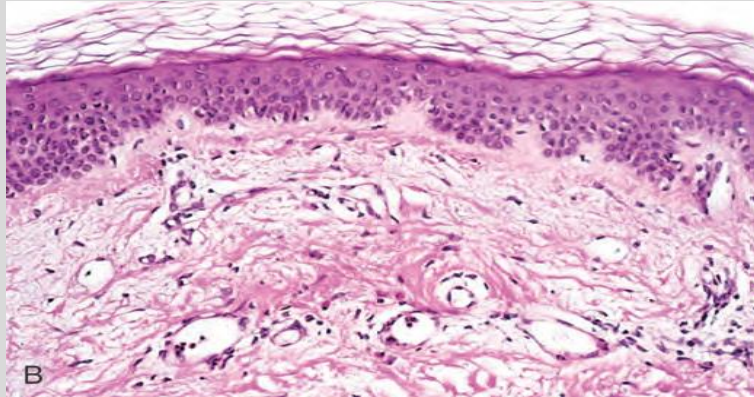
- **Localized mast cell degranulation**
- **Histamine** release, activates H1 & H2 receptors → ↑ vascular permeability, vasodilation → **wheals** + **pruritis**
- Affect **all ages**
- Commonly on trunk, extremities, face
- Majority of patients reveal **no underlying cause**
- Lesions develop & fade → hours/a day
- Can recur → **cycles** can last days/months



Urticaria: Pathogenesis

- **Mast cell-dependent, IgE-dependent:** Post-exposure to antigens → **Type I** Hypersensitivity reaction
 - When systemic → **anaphylaxis**
- **Mast cell-dependent, IgE-independent:** Substances **directly trigger mast cell** degranulation (examples: Opiates, certain antibiotics, contrast media)
- **Mast cell-independent, IgE-independent: Local factors** that ↑ vascular permeability w/o degranulating mast cells
 - **Aspirin** mechanism is unknown
 - **Hereditary Angioneurotic edema:** Rare, potentially life threatening genetic mutation w/ **deficiency of C1 esterase inhibitor** → ↑bradykinin (vasoactive mediator)

Urticaria: Histopathology



- **Dermal edema** - spacing of dermal collagen bundles
- Sparse superficial perivenular mononuclear cells, possibly eosinophils & neutrophils
- **Dermal lymphatics dilated** - edema fluid
- **No changes in epidermis**

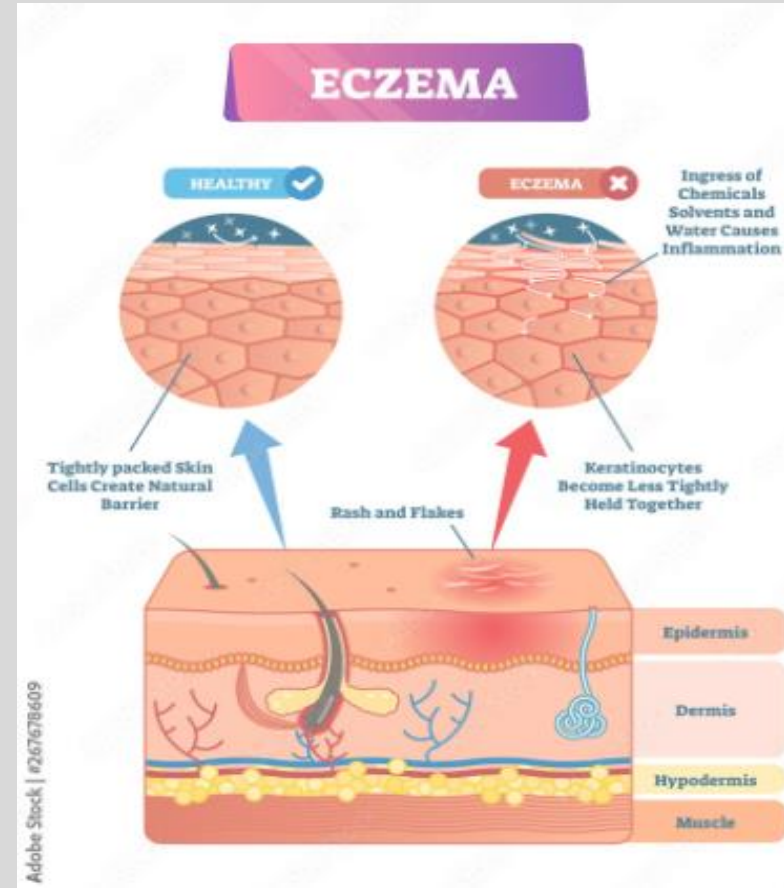
Acute eczematous dermatitis

- Inflammatory skin conditions w/ **erythema, edema, vesiculation & pruritis**
- **Type IV** hypersensitivity: **T-cell mediated** inflammation → **epidermal spongiosis** (edema)
- Types based on initiating factors:
 - **Atopic dermatitis**
 - **Allergic contact dermatitis**
 - **Primary irritant contact dermatitis**
 - **Photo-eczematous dermatitis**
 - **Drug-related eczematous dermatitis**



Robbins and Cotran, Atlas Pathology, 4th edition, 2021, Fig. 16.63

<https://www.langfordallergy.com/wp-content/uploads/2023/09/Elbow-with-Eczema-1536x1022.jpeg>

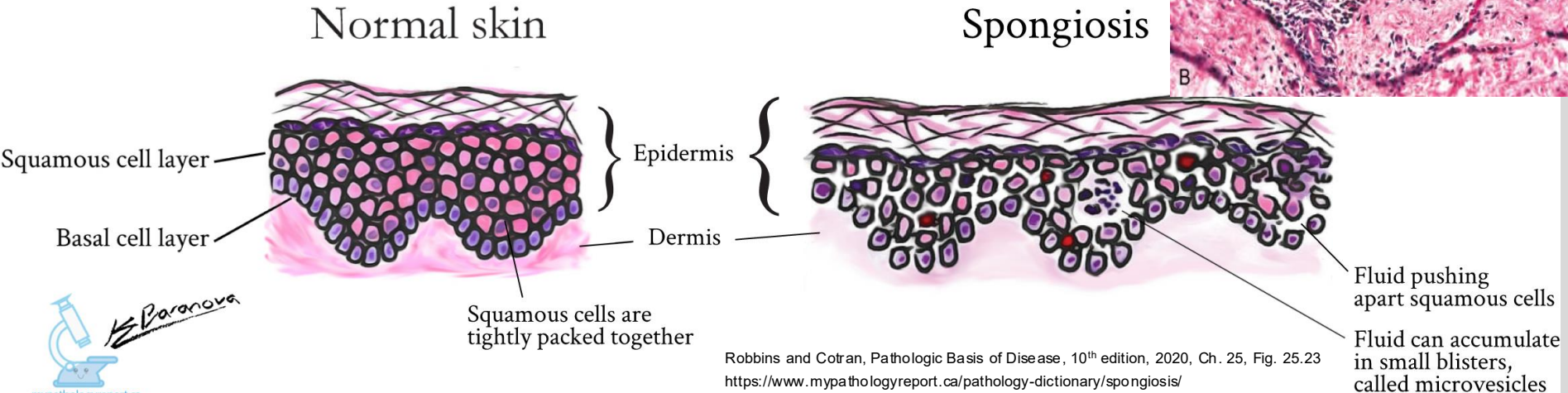
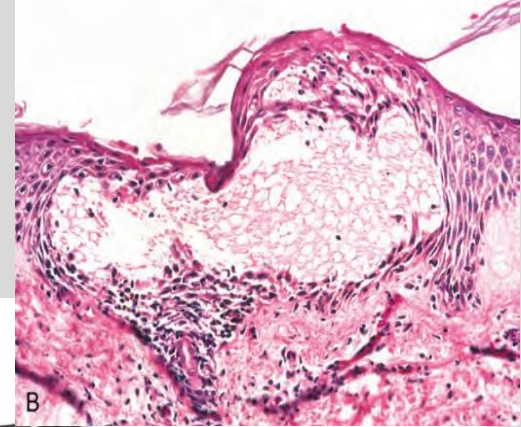


Ecematous dermatitis: Histopathology

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- **Spongiosis**
 - **Epidermal** intercellular edema (**vs. dermis** in urticaria)
 - Separates keratinocytes in stratum spinosum
- **Intra-epidermal vesicles**
 - Mechanical sheering (**scratching**) of desmosomes & cell membranes → progressive intercellular **fluid accumulation**
- Superficial, perivascular lymphocytic infiltrates



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- The diagram illustrates the pathogenesis of Atopic Dermatitis (AD) as a cycle involving skin barrier dysfunction, immune dysregulation, and inflammation.
- Barrier Dysfunction:** Triggered by *S. aureus* colonization, scratches, and allergens. This leads to decreased levels of *FLG* (Filaggrin) and *OVOL 1* (Ovolin-1).
 - Immune Response:**
 - Th2 Cells:** Central players that secrete *IL-4*, *IL-13*, and *IL-31*. They are also influenced by *OX-40L* and *CCL17*.
 - Th22 Cells:** Secrete *IL-4* and *IL-13*.
 - Th17 Cells:** Secrete *IL-13*.
 - Th1 Cells:** Present in the skin.
 - ILC2s (Innate Lymphoid Cells 2):** Secrete *IL-13* and *IL-5*. They are influenced by *IL-33* and *TSLP* (Thymic Stromal Lymphopoietin).
 - Inflammation and Itch:**
 - IL-5* recruits **Eosinophils**.
 - IL-4* and *IL-13* stimulate **Mast cells** to release **IgE**.
 - IL-31* and *IL-13* contribute to the sensation of **Itch**.

<https://onlinelibrary.wiley.com/doi/10.1111/1346-8138.15664>

Allergic contact dermatitis

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- **Type IV** hypersensitivity
- Triggered by **contact allergens** (ex. nickel, poison ivy, fragrances, latex)
- Affects **superficial dermis**, mononuclear inflammatory reaction w/o eosinophils



- Dermatology: An Illustrated Colour Text, Sixth Edition
- **Watchband** (clasp's metal alloy) etiology

Additional causes of eczema

- **Irritant contact dermatitis**
 - Direct chemical irritation (detergents, acids) produce **non-immunologic**, dose-dependent reaction
- **Photo-eczematous dermatitis**
 - UV light, double-edged sword for eczema
 - **Trigger for flares**, excessive exposure can damage the skin barrier
 - **Rx** in controlled doses - (UVB) reduces inflammation



Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fig. 25.23

Dermatology: An Illustrated Colour Text, Sixth Edition

Erythema multiforme: Overview

- Rare, **self-limiting** hypersensitivity
 - Reaction to certain **infections & drugs**
- Affects **all ages**
- Mediated by **T-cells** (both CD4+ and CD8+ T cells) → **Type IV** delayed reaction
- Triggers
 - **HSV** (most common), Mycoplasma pneumoniae, EBV, CMV, VZV, histoplasmosis
 - Sulfa drugs (**TMP-SMX**), NSAIDs, penicillins, barbiturates, phenytoin
 - Malignancies, autoimmune disease (SLE, IBD)

Erythema multiforme: Clinical

- **Multiple** lesions - **<10%** of total body surface area
- **Targetoid** – 3 concentric rings w/ dark, dusky center (**epidermal necrosis**)
- Symmetrical distribution over the body (hands, feet, limbs - acral distribution), can spread to face & trunk
- Often affects **mucosa**

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Source: K. Wolff, R.A. Johnson, A.P. Saavedra, E.K. Roth
Fitzpatrick's Color Atlas and Synopsis of Clinical
Dermatology, Eighth Edition. www.accessmedicine.com
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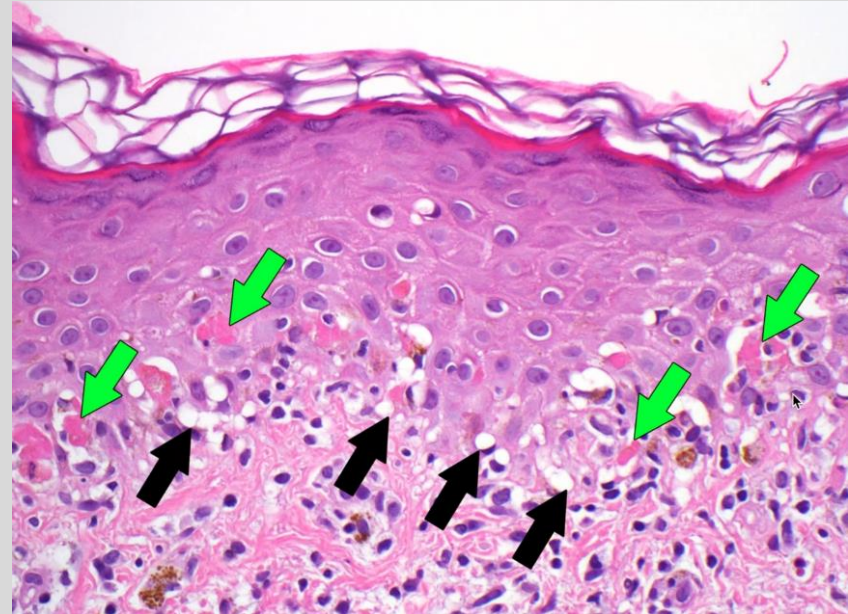
<https://dermnetnz.org/topics/erythema-multiforme>

Erythema multiforme: Histopathology

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- **Interface dermatitis** (dermo-epidermal junction pathology)
- Can see sloughing and blistering (aka separation from underlying tissue)



<https://www.youtube.com/watch?v=HyOxbBmNtfw>

Stevens-Johnson Syndrome (SJS) & TEN

- **Severe mucocutaneous** reaction
 - **Widespread epidermal necrosis & detachment**
- Often triggered by **meds or infections** (ex. **sulfa** drugs most commonly)
- Clinically: Lesions involve skin and lips/oral **mucosa**, conjunctiva, urethra, genital areas
 - Initially erythematous macules → bullae → epidermal detachment (**+ Nikolsky sign**)
 - **>10% skin** involvement
 - If **>30%**, renamed **Toxic Epidermal Necrolysis (TEN)**
- Management → Stop offending drug + supportive care
- Prognosis -> **5-10% mortality** for SJS, **>30% for TEN**



Chronic inflammatory dermatoses

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1. Psoriasis
2. Lichen Planus
3. Seborrheic dermatitis

Psoriasis: Overview

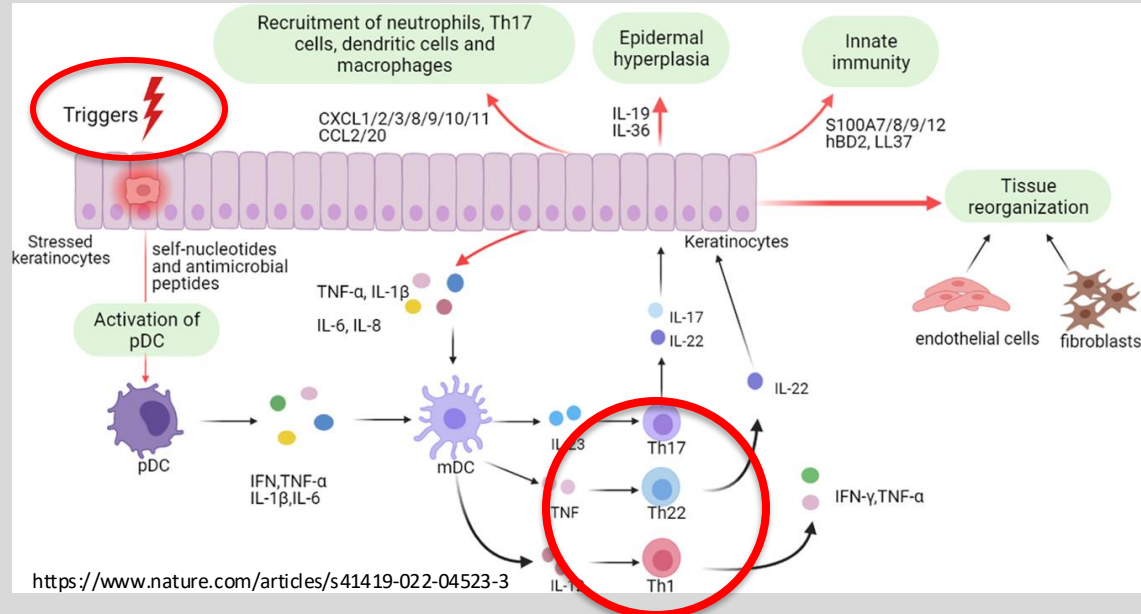
- Chronic, **autoimmune mediated** inflammatory skin disorder characterized by **hyperproliferation of keratinocytes**
- Typically affects the **extensor surfaces, groin, intergluteal, armpit**
- **Scaly** skin plaques
- Symptoms often start between ages 15 and 25 but can start at any age
- **Unknown trigger** activates innate immunity
- 15% have associated **psoriatic arthritis**

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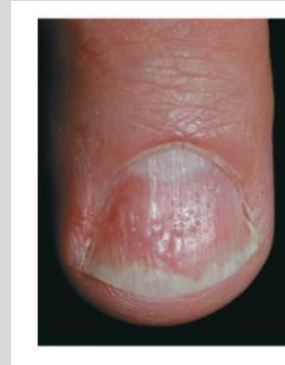


Robbins and Cotran, Pathologic Basis of Disease, 10th edition, 2020, Fib. 25.25



Psoriasis: Clinical

- Lesion appearance varies: Typically well-demarcated, pink to **salmon plaques** covered by loosely adherent **silver-white scale**
- **Nail changes**
- **Koebner Phenomenon**
- **Auspitz sign**



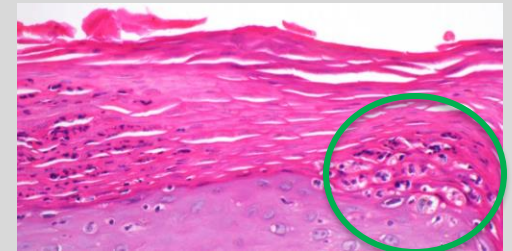
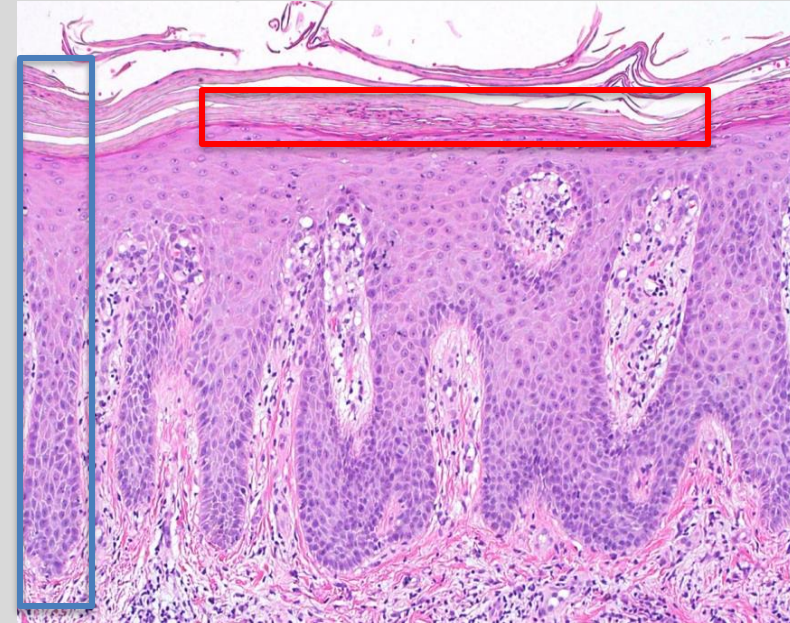
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Psoriasis: Histopathology

- **Acanthosis:** Marked epidermal thickening
- **Elongation of the rete ridges** → test tubes (blue box)
- **Parakeratosis (red box)**
 - Retention of nuclei in stratum corneum
- **Munro microabscesses (green circle)**
 - Neutrophils form small aggregates within superficial epidermis (spongiform pustules)
- **Psoriatic Plaques**
 - Thinning of the section of epidermis overlaying tips of dermal papilla



Lichen Planus: Overview

- **Pruritic, purple, polygonal, planar, papules, plaques** that affects skin, mucosa, nails, hair follicles
- Typically **middle aged**
- **Self-limiting** → resolves spontaneously over course of a few years
- Pathogenesis not fully known
 - Possibly CD8+ T-cell response to antigens in basal epidermal or dermo-epidermal junction
- Associated w/ **Hepatitis C**, although **most** cases **idiopathic**

Lichen Planus: Clinical

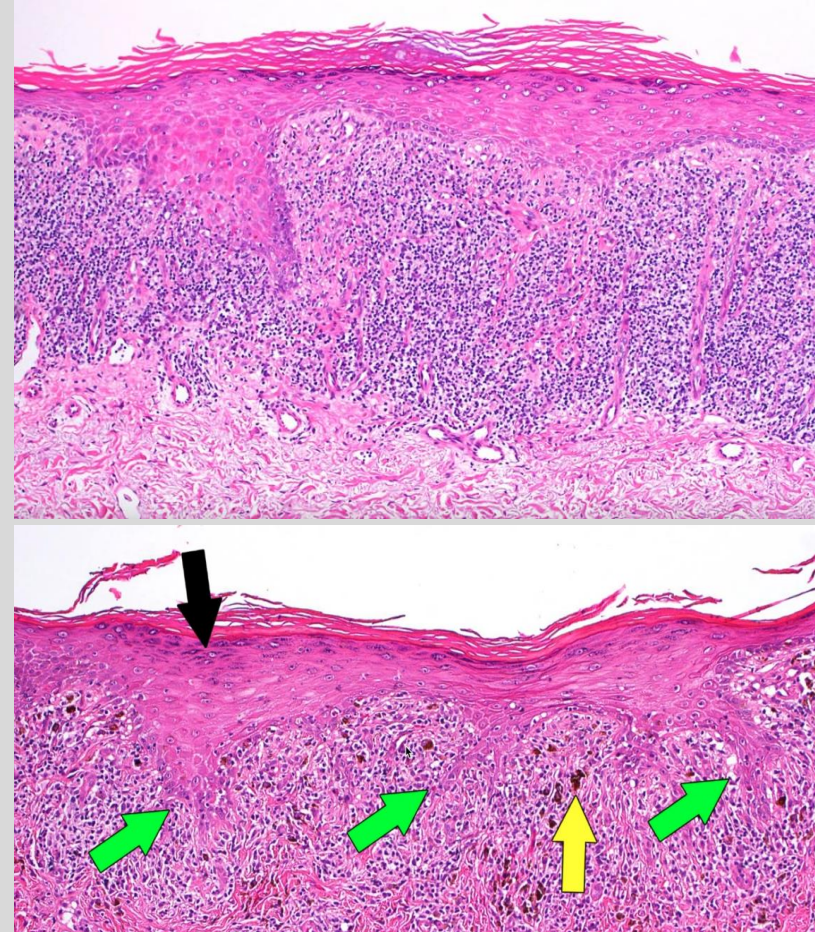
- Itchy, **violaceous**, flat-topped papules or plaques
- **Wickham Striae** → white dots or reticulated lines in the oral mucosa
- Typically **symmetrical**: **Wrists & elbows**



Source: K. Wolff, R.A. Johnson, A.P. Saavedra, E.K. Roh:
Fitzpatrick's Color Atlas and Synopsis of Clinical
Dermatology, Eighth Edition: www.accessmedicine.com
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Lichen Planus: Histopathology

- **Interface dermatitis** (dense, band-like, lymphocytic infiltrate at the dermal-epidermal junction)
- **Sawtoothing** of the rete ridges (green arrows)
- **Civatte bodies**: Anucleate, necrotic basal keratinocytes
- **Hypergranulosis/Hyperkeratosis**: Thickening of the granular layer & stratum corneum (black arrow)
- **Pigment falling into dermis** (yellow arrow)



Seborrheic dermatitis: Overview

- Common ~5% of general population
- Chronic, relapsing inflammatory skin disorder affecting **sebaceous gland-rich areas** (ex. **scalp & forehead**)
- Associated with **epidermis inflammation** (but not a disease of sebaceous glands themselves)
- **Cause is unknown**
 - ↑ **sebum production**
 - Overgrowth of ***Malassezia*** yeast is associated
- Clinically: **Scaly, greasy, yellow plaques** on scalp, face, chest
 - **Dandruff** is a common clinical expression of scalp seborrheic dermatitis
- Histopathology: Acute, subacute or chronic **spongiotic dermatitis**
 - Spongiosis with **overlying scale crust (parakeratosis** containing **neutrophils & serum** at hair follicles ostia)
- Rx: Antifungals, low strength steroids, medicated shampoos (selenium sulfide, zinc pyrithione)

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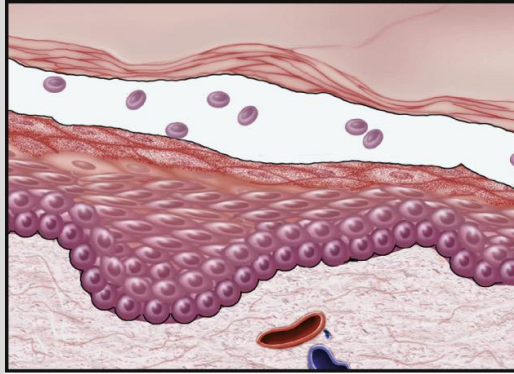


<https://www.gponline.com/seborrheic-dermatitis-clinical-review/dermatology/dermatology/article/1181296>

Bullous (blistering) disorders

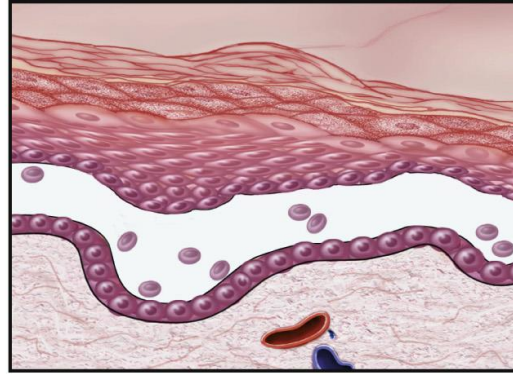
1. Pemphigus Vulgaris
2. Bullous pemphigoid
3. Dermatitis herpetiformis
4. Epidermolysis bullosa

Type of blister helps determine the cause



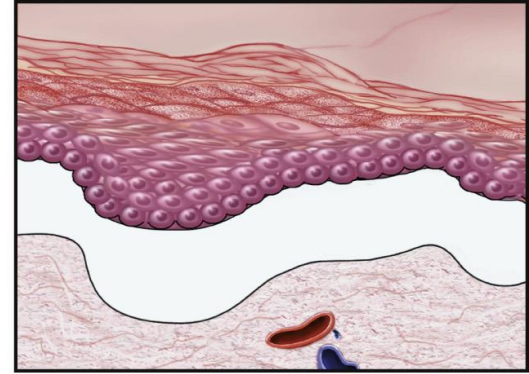
A Subcorneal

Stratum corneum forms the roof of the bulla (ex. friction blister, pemphigus foliaceus)



B Suprabasal

A portion of epidermis, including the stratum corneum, forms the roof (ex. **pemphigus vulgaris**)



C Subepidermal

Entire epidermis separates from the dermis (ex. **bullous pemphigoid**)

Pemphigus vulgaris: Clinical findings

- Inflammatory, blistering skin diseases caused by **autoantibodies (IgG)**
- Auto-abs **disrupt intercellular keratinocyte attachments** within epidermis & mucosal epithelium
- Patients in **30s-50s** & males, females equally effected
- Usually **benign**, but extreme variants **can be fatal** w/o Rx
- Involved **skin & mucosa** (**oral** ulcers may present months **before skin** lesions)
- Lesions are superficial vesicles and blisters rupture easily
 - (+ **Nikolsky sign**)
- Treat with immunosuppressive agents → ↓ Abs

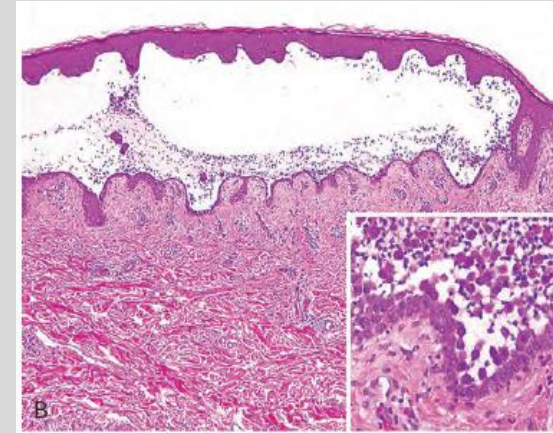
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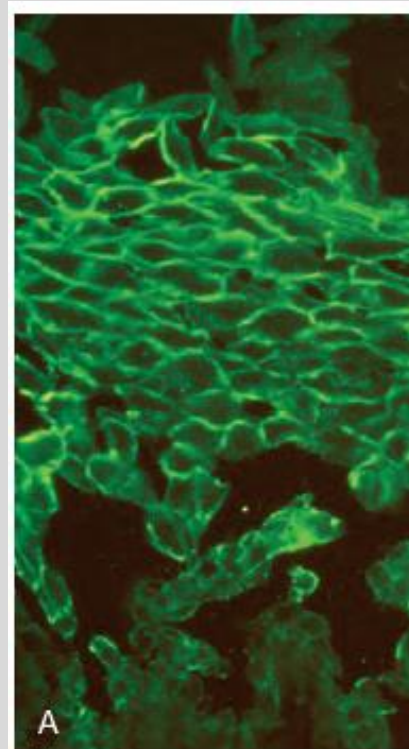
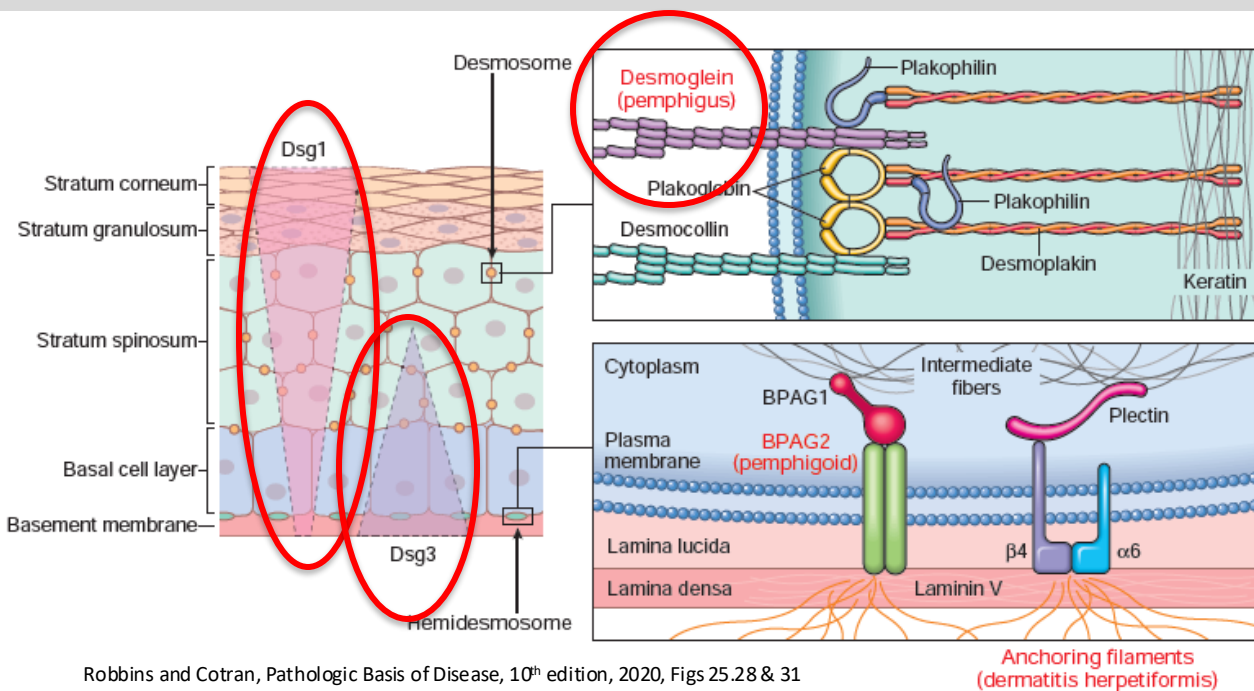
Pemphigus vulgaris: Histopathology

- Autoimmune: **IgG** autoantibodies against **desmogleins** (subunit of desmosome)
- Causes **suprabasal** blisters
- **Acantholysis**: Dissolution of intercellular bridges immediately above basal cell layer
- **Row of tombstones**: Blister leaves behind single layer of intact basal cells



Pemphigus vulgaris: Histopathology - direct immunofluorescent microscopy detects epidermal autoantibodies

- Distinct **net-like** pattern of intercellular IgG deposits
- IgG is usually seen at all levels of the epithelium
- Distribution of **desmoglein 1 & 3**



Bullous pemphigoid: Clinical findings

- Inflammatory, blistering skin diseases caused by **autoantibodies (IgG)**
- Usually older/**elderly** patients
- Inner thigh, flexor surfaces, axillae, groin affected
- Oral lesions less likely than pemphigus vulgaris (**oral** lesions usually appear **after skin** lesions)
- **Tense bullae** that do not rupture easily
 - (**Negative Nikolsky sign**)
- Also treat w/ immunosuppressive agents

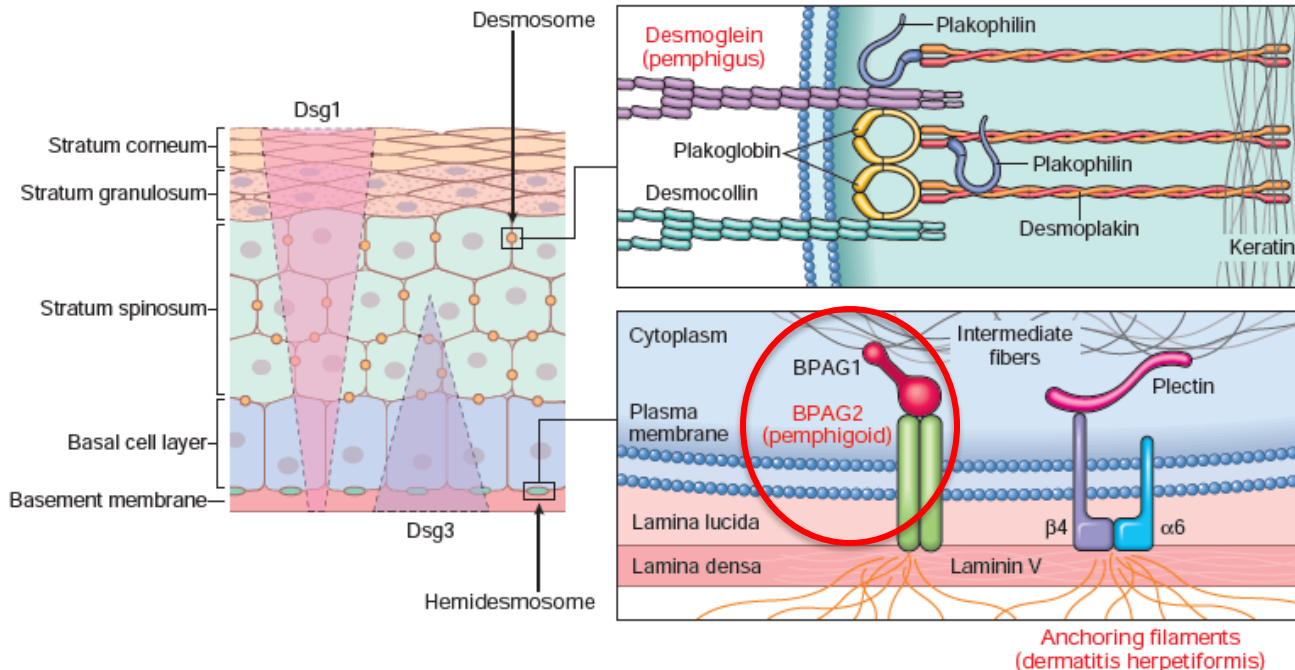


Bullous pemphigoid: Pathogenesis

- **IgG autoantibodies** against **BPAG2** - part of **hemidesmosomes** (attach basal keratinocytes to the basement membrane)
- Forms **subepidermal blisters** (lamina lucida), nonacantholytic

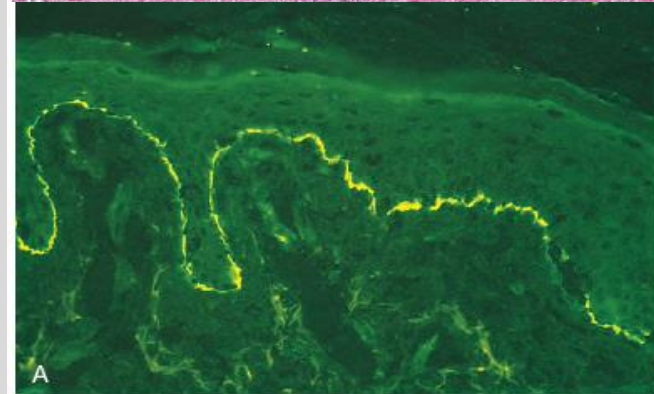
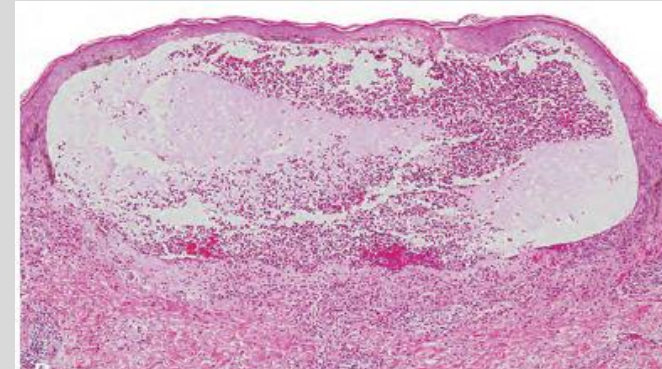
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Bullous pemphigoid: Histopathology

- **Sub-epidermal & non-acantholytic**
- **Superficial dermal edema**
- Basal layer vacuolization, leading to fluid-filled blisters
- Full thickness of epidermis serving as roof of blister
 - - **Nikolsky sign**
- Direct immunofluorescent microscopy detects **subepidermal autoantibodies** - linear, continuous IgG deposition along the dermo-epidermal junction (hemidesmosomes are here)

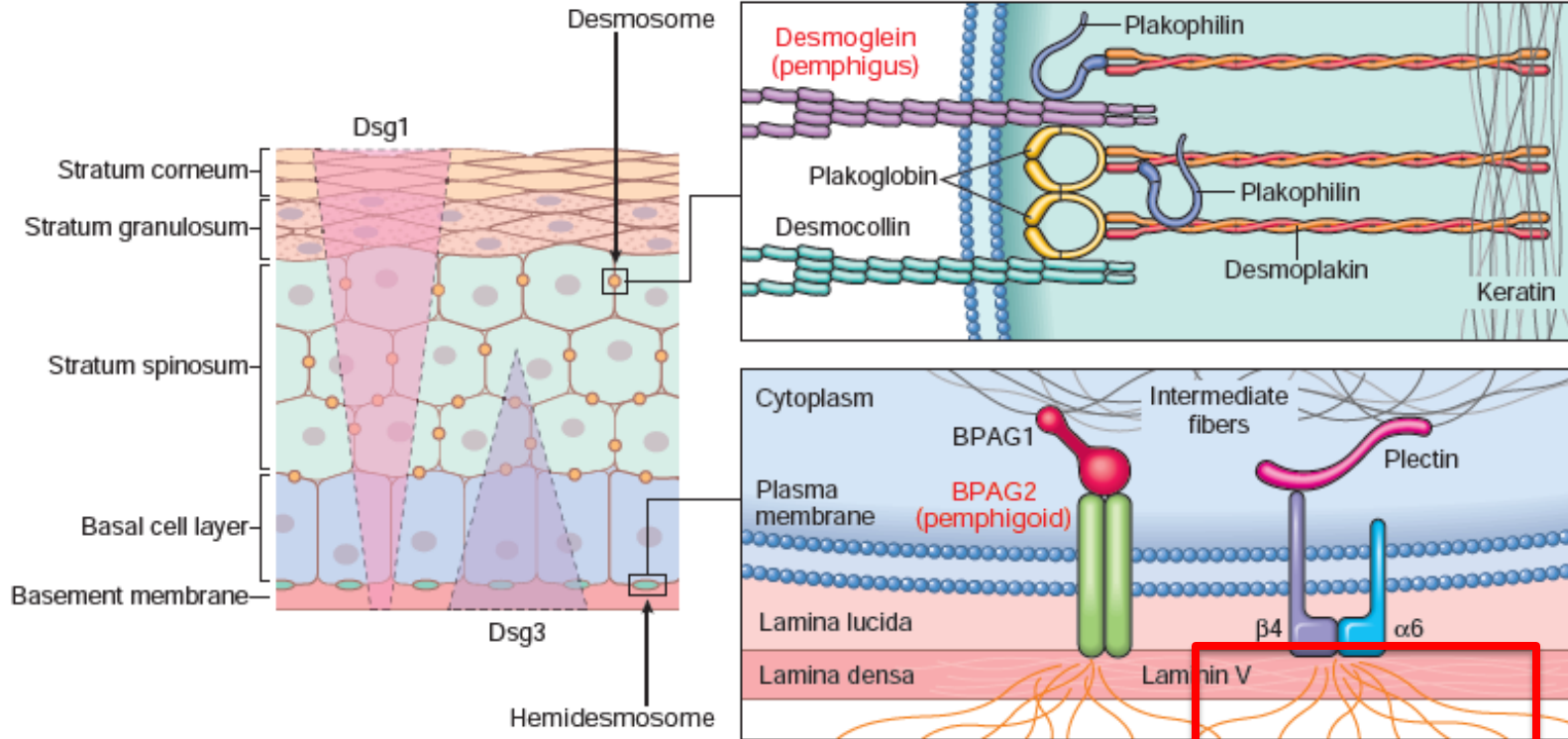


Dermatitis Herpetiformis: Clinical findings

- Rare skin disorder w/ urticaria and **grouped vesicles**
- Affects mainly males in 30s-40s
- Association with **celiac disease** (**IgA antibodies** to gluten → cross react with **reticulin**)
- **Subepidermal** blister
- Very **pruritic**
- Symmetrical lesions, affecting extensor surfaces (knees/elbows), back and buttocks

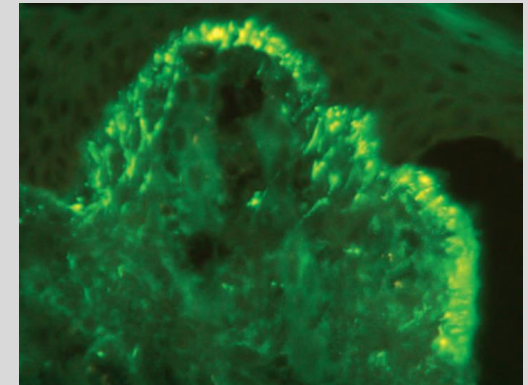
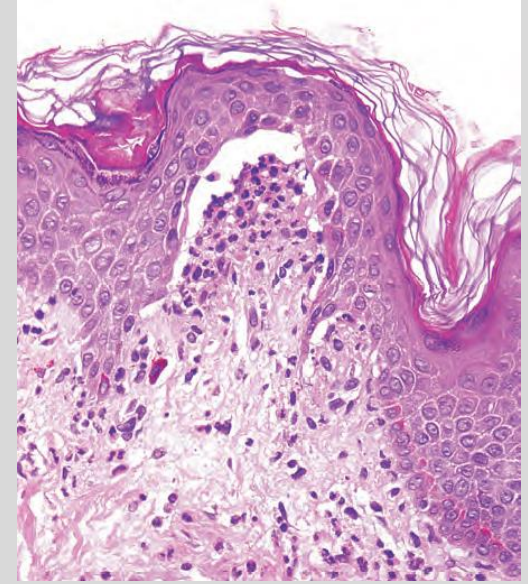
<https://revivalresearch.org/blogs/dermatitis-herpetiformis-vs-eczema/>





Dermatitis herpetiformis: Histopathology

- **Microabscesses:** Fibrin and **neutrophils** accumulate at **tips of dermal papillae**
- Basal cell layer vacuolization and dermo-epidermal junction separation → **subepidermal blisters**
- Direct immunofluorescent microscopy detects a **discontinuous, granular IgA** deposits at tips of dermal papillae (anti-tissue transglutaminase (tTG) and anti-epidermal transglutaminase (eTG))



Epidermolysis bullosa: Overview

- **Genetic** blistering disorder
 - Defective epidermal-dermal adhesion
- Wide range of phenotypes produces multiple subtypes
 - Determined by which epidermal or dermal protein is affected: Keratin, laminin, collagen, etc.
- **Fragile skin** that blisters from **minor trauma**
 - Mucosal involvement in severe forms
- Common feature - blisters at sites of pressure, rubbing, or trauma, at or **soon after birth**
- Severe forms may cause contractures
 - Patients lose use of their hands and feet
- Wound care and infection prevention critical

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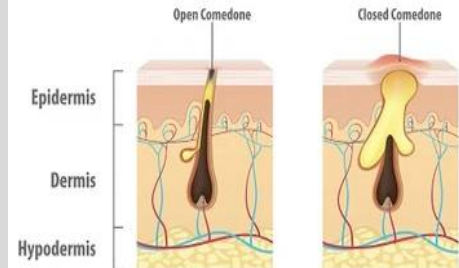


Acne vulgaris: Clinical findings

- Virtually universal in the middle to late teenage years
- Incidence: M=F, males typically more severe
- Exacerbated by
 - Drugs – **corticosteroids**, **testosterone**, gonadotropins, **contraceptives**
 - **Occlusion** of sebaceous glands – heavy clothing, cosmetics, chin straps in football
- Can be familial - possibly hereditary
- **Noninflammatory** acne - both types can coexist
 - **Open comedones** (blackheads): Central black **keratin plug** - oxidation of melanin
 - **Closed comedones** (whiteheads): Keratin plug is trapped beneath the epidermal surface
- **Inflammatory** acne - both types can coexist
 - **Erythematous** papules, nodules, pustules
 - **Severe variants** can result in sinus tract formation and **scarring**
- Histopathology shows variable infiltrates of lymphocytes and macrophages present in and around follicles



OPEN COMEDONE & CLOSED COMEDONE



Acne vulgaris: Pathogenesis, treatments

- Likely multifactorial causes
 1. Keratinization of lower portion of follicular stalk and development of **keratin plug** - that blocks outflow of sebum to the skin surface
 2. Hypertrophy of sebaceous glands - under the influence of **androgens** during **puberty**
 3. Lipase-synthesizing bacteria (**Propionibacterium acnes**) colonize hair follicle converting lipids within sebum to proinflammatory fatty acids
 4. Secondary inflammation of involved follicle
- Rx: Antibiotics for elimination of bacteria & isotretinoin has strong anti-sebaceous action

Rosacea: Overview

- Common in **middle age and older (females** more often)
- 4 Stages
 - Flushing episodes
 - Persistent erythema and telangiectasia
 - Pustules and papules
 - **Rhinophyma** (permanent thickening of nasal skin by confluent erythematous papules)
- Can have **ocular** involvement
- **Cause unknown:** However there is a high cutaneous levels of **cathelicidin**, an antimicrobial peptide involved in innate immunity against bacteria, fungi, parasite and enveloped viruses

<https://www.rosacea.org/patients/all-about-rosacea>



FACIAL REDNESS

Flushing and persistent redness. Visible blood vessels may also appear.



BUMPS AND PIMPLES

Persistent facial redness with papules and pustules (bumps and pimples.)



SKIN THICKENING

Skin thickening and enlargement (phymatous changes), usually around the nose.



EYE IRRITATION

Watery or bloodshot appearance, irritation, burning or stinging. Also known as ocular rosacea.

Erythema nodosum: Clinical findings

- **Panniculitis:** Inflammatory reaction in **subcutaneous adipose tissue**
- Mostly involves lower **legs & shins**
- Raised, poorly defined erythematous, **painful** nodules and plaques
- Triggers
 - Infections – beta-hemolytic strep (GAS/GBS), **TB**
 - Drugs – sulfas and **contraceptives**
 - **Sarcoidosis, IBD, Neoplasms**
- Can be accompanied by systemic signs and symptoms including **fever & malaise**
- Treat underlying condition, but usually resolves on its own or anti-inflammatory medications may be used if painful

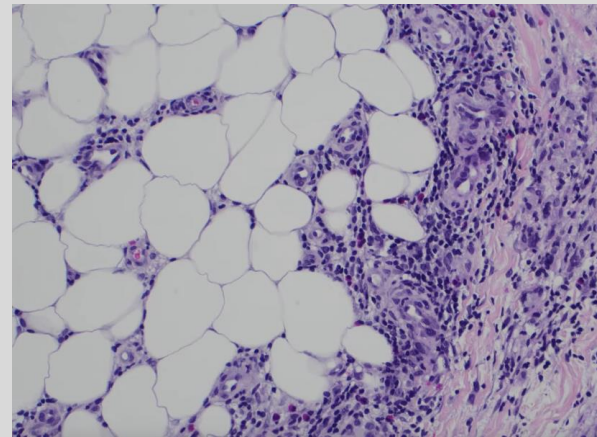
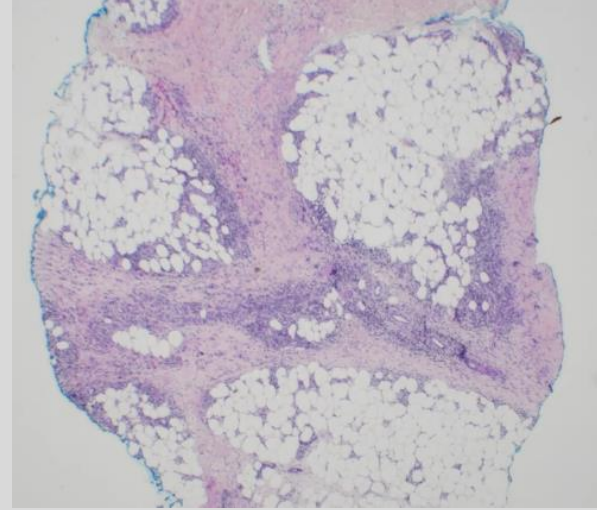


<https://mdsearchlight.com/health/erythema-nodosum/>

Erythema nodosum: Histopathology

- Punch **biopsy** used for diagnosis:
 - Thickened septa between fat globules
 - Fibrosis in the septa
 - Infiltration by lymphocytes, histiocytes, occasional eosinophils

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Lecture and Lab Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>